ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C5 · SOUND

Making and hearing sounds

★☆☆
Grade 5

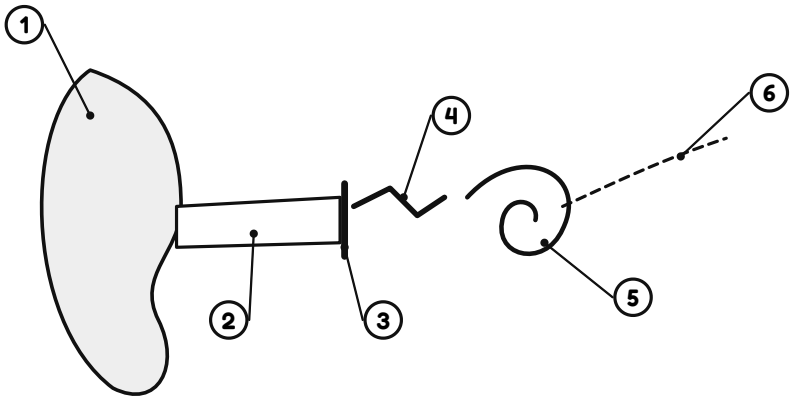
Learning goals

- Recall how sound is made and heard.
- Label the ear.

1 Label the parts of the ear. [6]

LABELS

eardrum **nerve to the brain** **ear canal** **tiny bones** **outer ear** **cochlea**



(1) outer ear (2) ear canal (3) eardrum

(4) tiny bones (5) cochlea (6) nerve to the brain

2 Fill in the gaps. [5]

WORD BANK

low **vibrations** **high** **wave** **space** **gases** **light**

All sounds are made by vibrations. Sound travels as a wave. It can travel through solids, liquids and gases, but not through empty space. Fast vibrations make a high pitch.

3 What vibrates to make the sound? [4]

Instrument	What vibrates?
dhol (drum)	the drum skin
ektara	the string
bansuri (flute)	the air inside
your voice	the vocal cords in your throat

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. Sound travels faster through steel than through air.	✔		
2. A bigger vibration makes a quieter sound.		✔	A bigger vibration makes a louder sound.
3. Astronauts on the Moon can hear each other without radios.		✔	There is no air on the Moon, so sound cannot travel.
4. A shorter string makes a higher pitch.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C5 · SOUND

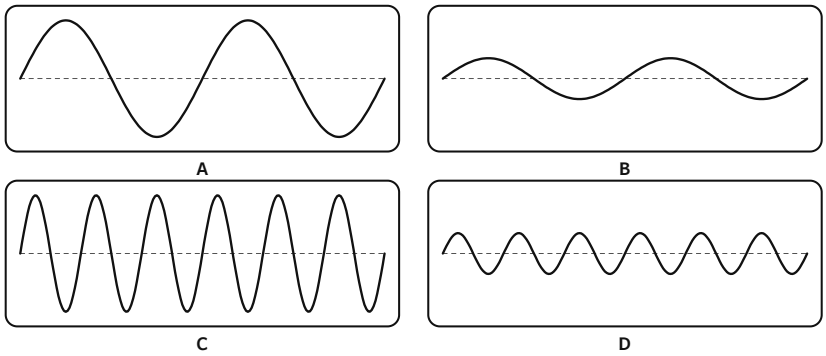
Pitch, loudness and waves

★★☆
Grade 5

Learning goals

- Use wave diagrams.
- Explain pitch and loudness changes.

1 **Look at the four waves A to D.** [4]



a) Which wave is the loudest with the lowest pitch?

A

b) Which wave is the quietest with the highest pitch?

D

c) What is the same about waves A and C? What is different?

Same height, so the same loudness; C has more waves close together, so a higher pitch.

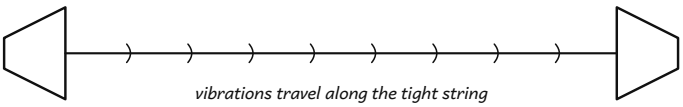
2 **An ektara player squeezes the bamboo neck to make the string tighter. What happens to the pitch? Why?** [2]

The pitch gets higher, because a tighter string vibrates faster.

3 **Why do we see lightning before we hear the thunder?** [2]

Light travels much faster than sound, so it reaches us first.

4 **A string telephone works only when the string is tight. Explain.** [2]



vibrations travel along the tight string

Vibrations travel along a tight string from one cup to the other; a loose string does not pass the vibrations on.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C5 · SOUND

Ruler twang: length and pitch

★★☆
Grade 5

Learning goals

- Investigate pitch with a fair test.
- Plot and use a line graph.
- Explain using vibrations.

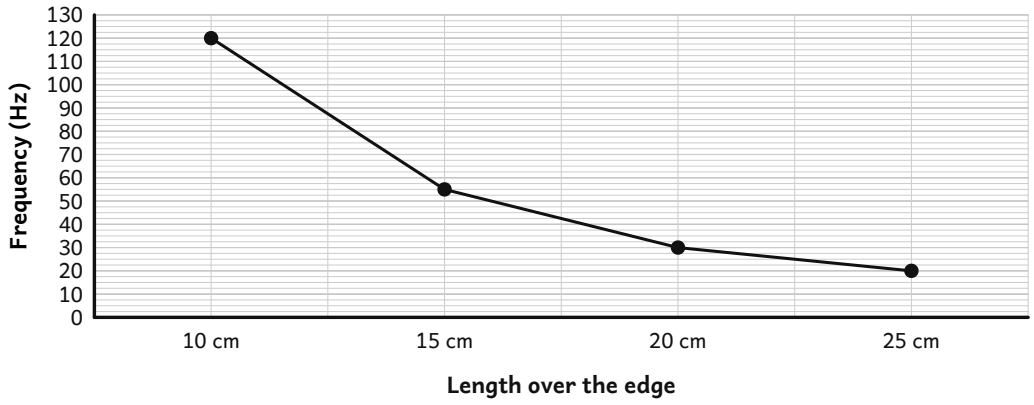
Tania held a ruler on the edge of a table and twanged the end. She changed how much of the ruler stuck out over the edge. A sound app on a tablet counted the vibrations each second (frequency).

1 **Name the variables.** [3]

Change	length of ruler over the edge
Measure	frequency (vibrations per second, Hz)
Keep the same	the same ruler, how hard she pressed and twanged, same table

2 **Plot the results as a line graph.** [4]

Length over the edge (cm)	10	15	20	25
Frequency (Hz)	120	55	30	20



3 **Describe the pattern. Which length made the highest pitch?** [2]

The longer the ruler over the edge, the lower the frequency and pitch. 10 cm made the highest pitch.

4 **Predict the frequency for 12 cm.** [1]

between 55 and 120 Hz, about 85–95 Hz

5 **How could Tania test how loudness changes? Write a plan in one sentence.** [2]

Keep the length the same and twang the ruler with a small and a big push, measuring the loudness (dB) with the app.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C5 · SOUND

Too loud! Noise in Dhaka

★★★
Grade 5

Learning goals

- Use data about loudness.
- Explain health effects of noise.
- Evaluate solutions from different viewpoints.

Science | Health | Maths | Citizenship | Ethics

In 2022, a United Nations Environment Programme report named **Dhaka** as one of the noisiest cities in the world. Car and bus horns, construction machines, loudspeakers and generators fill the streets with sound.

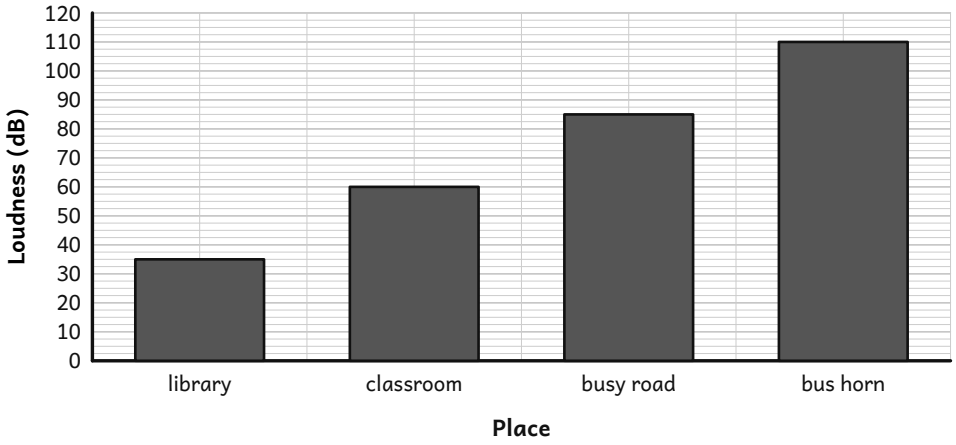
Doctors say that living with loud noise every day can cause **hearing loss**, headaches, stress, high blood pressure and poor sleep. Children in noisy schools find it harder to concentrate and learn.

Bangladesh has **noise rules**: areas around hospitals, schools and offices are meant to be **silent zones**, with limits as low as 50 dB in the daytime, and very loud hydraulic horns are banned. But the rules are often broken.

Sources: UNEP Frontiers Report 2022; Bangladesh Noise Pollution (Control) Rules 2006.

1 **A class measured the loudness at different places with a sound meter app (example readings).** [4]

Place	school library	classroom	busy road outside school	bus horn nearby
Loudness (dB)	35	60	85	110



a) Draw the bar chart.

b) Which places are louder than the 50 dB limit for a silent zone?

classroom (60), busy road (85), bus horn (110)

c) Which place could damage hearing if people stayed there for a long time?

the busy road and near bus horns (85 dB and above)

2 **Explain why a loud horn could harm the ear, using what you know about the eardrum.** [2]

Very big vibrations make the eardrum and the tiny parts of the cochlea vibrate very strongly, which can damage them.

3 **What might each person say?** [3]

Person	View
a bus driver	Everyone pushes in; I must honk to stay safe and keep moving.
a teacher near the road	Children cannot hear me or concentrate; the noise hurts learning.
a traffic police officer	Horn rules are hard to enforce; drivers need training and checks.

4 **Design a "Quiet School Zone" plan with three actions.** [3]

e.g. "No horn" signs and speed bumps near school; plant trees and build walls to block sound; awareness campaign for drivers; fines for hydraulic horns; double windows or curtains in classrooms.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C5 · SOUND

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

H	E	U	A	Y	S	T	R	I	N	G
A	C	B	T	L	O	U	D	Y	C	P
D	H	N	S	R	R	F	L	S	V	E
J	O	Y	F	V	M	G	T	V	E	F
K	Y	J	V	L	F	I	D	F	A	Y
W	J	F	I	D	U	T	E	F	R	C
A	W	J	B	R	U	P	C	I	D	C
V	O	S	R	U	F	I	I	B	R	G
E	M	O	A	M	K	T	B	A	U	Y
A	U	M	T	I	S	C	E	A	M	I
J	A	E	E	N	I	H	L	K	O	G

decibel **drum** **eardrum** **echo** **loud** **pitch** **string** **vibrate** **wave**

2 **Which change makes a guitar string sound higher?** [1]

- ○ make it longer
- ○ make it looser
- ● **make it shorter**
- ○ pluck it more gently

3 **Why can't sound travel in space?** [1]

- ○ it is too cold
- ● **there are no particles to pass on vibrations**
- ○ it is too dark
- ○ the Sun is too loud

4 **Draw two waves: one quiet, high-pitched sound and one loud, low-pitched sound.** [2]

quiet, high pitch

Short waves, close together.

loud, low pitch

Tall waves, far apart.

5 **Challenge. Whales can hear each other across hundreds of kilometres of ocean. Use what you know about sound to suggest why.** [2]

Sound travels well and fast through water because the particles are close together; low-pitched whale calls travel very far.